NOTE: Image files are stored separately in the repository and may not render inline in the GitHub notebook preview. For full image rendering please view the included PDF export (ForestGreenFinalFile.pdf), and for interactive graphs please open locally in Jupyter notebook.

Group Forest Green

Participants: Trym Stuen, Anna Kilvington and Téa Leela Banerjee-Wøien

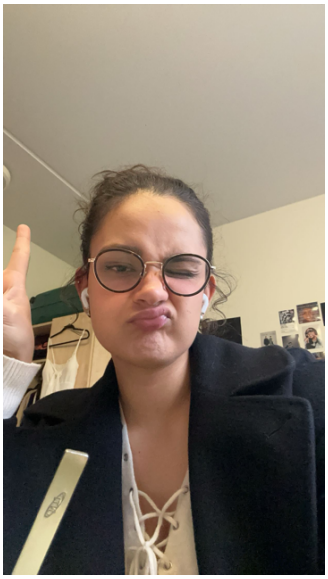
Trend Analysis on Protected Areas (2013-2023)



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Project Illustration (Yosie, 2024)

Course Information

Course: Data Science (9101)
Semester: Autumn 2026
Institution: Institut Barcelona d'Estudis Internacionals (IBEI)
Program: International Relations Master **Instructor:** Prof. Navid Hashemi

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1. Introduction

1.1 Project Objective

This project's objective is to identify patterns of influence on protected areas around the world using data from 2013 to 2023. Environmental quality is a growing concern in the modern world, making protected areas an imperative measure for the perseverance of biodiversity. Not only do regulated areas serve as a protective barrier for ecosystems, it also plays a large economic bi-role in nature-based tourism and to the intangible, psychological factors of societal well-being and feeling of unity (Harmon, 2004, p. 9).

The aim of this project is to recognise globally generalisable factors that potentially influence the frequency at which nations across the world prioritise protected areas within their borders. Identifying possible risk-factors could contribute to the development of more comprehensive frameworks for protective areas, potentially assisting the UN's 13th Sustainable Development Goal (SDG); Climate Action. By using large datasets from the World Bank we will scan for patterns and variables that predict outcomes related to protective areas, contributing to the development of protective schemes likely to yield environmental, economic, and psychological benefits.

1.2 Research Question and Hypothesis

Our analysis will be guided by the following questions:

R1: Is there a relationship between the quantity of protected areas and equal gender employment?

R2: Is there a relationship between women's rights performance and the quantity of protected areas?

R3: Do increased levels of corruption limit environmental protection?

R4: Does political stability encourage more environmental protection?

R5: Does a natural resource based economy lead to less prevalence of protected areas?

Hypothesis
Our hypothesis is that countries with higher levels of political stability, labour gender equality, and with a less resource-rent based economy will perform better in frequency of protected areas (marine and terrestrial).

2. Dataset Description and Data Exploration

2.1 Data Description and Preparation

Before beginning project, the dataset had to be prepared for further analysis. We defined our specific indices:

Political Stability

Political Corruption Index

Female Labour Force Participation

Proportion of Seats Held by Women in National Parliament

Terrestrial and Marine Protected Areas

Mineral Rents

Total Natural Resources Rents

Our dataset contained a significant amount of irrelevant indices, therefore, we extracted our selected indicators separately from the World Bank site and combined them to create a custom table for representation.

We also removed regional and income groups aggregates to only assess the 217 individual countries and territories to narrow down the data to fit our analysis.

The code of how we accomplished this is provided below.

In [2]:
```
import pandas as pd
```

In [3]:
```
df = pd.read_csv('combined_wdi.csv')
```

In [ ]:
```

```

In [4]:
```
df
```

Out[4]:

	country_code	country	year	ER_PTD_TOTL_ZS	GOV_WGI_CC_E
0	ABW	Aruba	1986	NaN	N
1	ABW	Aruba	1987	NaN	N
2	ABW	Aruba	1988	NaN	N
3	ABW	Aruba	1989	NaN	N
4	ABW	Aruba	1990	NaN	N
...	...	...	...	...	
13127	ZWE	Zimbabwe	2021	27.2	-1.2119
13128	ZWE	Zimbabwe	2022	27.2	-1.1942
13129	ZWE	Zimbabwe	2023	28.3	-1.1985
13130	ZWE	Zimbabwe	2024	28.3	-1.269
13131	ZWE	Zimbabwe	2025	28.3	-1.292

13132 rows × 10 columns

In [5]:
```
df = pd.read_csv("combined_wdi_countries_only.csv")

df.head()        # first 5 rows
df.shape         # (10461, 10)
df.info()        # columns, types, non-null counts
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10461 entries, 0 to 10460
Data columns (total 10 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   country_code       10461 non-null  str    
 1   country            10461 non-null  str    
 2   year               10461 non-null  int64  
 3   ER_PTD_TOTL_ZS     2746 non-null   float64
 4   GOV_WGI_CC_EST     5408 non-null   float64
 5   GOV_WGI_PV_SC      5462 non-null   float64
 6   NY_GDP_MINR_RT_ZS  9267 non-null   float64
 7   NY_GDP_TOTL_RT_ZS  9235 non-null   float64
 8   SG_GEN_PARL_ZS     5259 non-null   float64
 9   SL_TLF_TOTL_FE_ZS  6718 non-null   float64
dtypes: float64(7), int64(1), str(2)
memory usage: 945.1 KB
```

Here we are ensuring the type of data we have is correct before starting: they are all floats and integers.

In [6]:
```
# Show every column and don't wrap the table
pd.set_option('display.width', 250)
pd.set_option('display.max_columns', None)

# World Bank codes for regions, income groups, and other aggregat
AGGREGATES = {
    'AFE','AFW','ARB','CEB','CSS','EAP','EAR','EAS','ECA','ECS','
    'FCV','HIC','HPC','IBD','IBT','IDA','IDB','IDX','LAC','LCN','
    'LMC','LMY','LTE','MEA','MIC','MNA','NAC','OED','OSS','PRE','
    'SAS','SSA','SSF','SST','TEA','TEC','TLA','TMN','TSA','TSS','
}

# Load the combined file and keep only countries and territories
df = pd.read_csv('combined_wdi.csv')
countries = df[~df['country_code'].isin(AGGREGATES)].reset_index(

# Print the table
print(f"\n{countries['country_code'].nunique()} countries/territo
df.head()
```

```
217 countries/territories when aggregates are removed, 10461 rows
```

Out[6]:

	country_code	country	year	ER_PTD_TOTL_ZS	GOV_WGI_CC_EST	G
0	ABW	Aruba	1986	NaN	NaN	
1	ABW	Aruba	1987	NaN	NaN	
2	ABW	Aruba	1988	NaN	NaN	
3	ABW	Aruba	1989	NaN	NaN	
4	ABW	Aruba	1990	NaN	NaN	

In [7]:
```
import numpy as np

# Load dataset
df = pd.read_csv('combined_wdi_countries_only.csv')

# Define custom variable labels
variable_descriptions = {
    "GOV_WGI_PV_SC": "Political Stability (SCORE 0-100)",
    "GOV_WGI_CC_EST": "Control of Corruption (governance estimate
    "SL_TLF_TOTL_FE_ZS": "Female Share of the Labour Force (%)",
    "SG_GEN_PARL_ZS": "Women's Share of Parliamentary Seats (%)",
    "ER_PTD_TOTL_ZS": "Protected Areas (% of total land and marir
    "NY_GDP_MINR_RT_ZS": "Mineral Rents (% of GDP)",
    "NY_GDP_TOTL_RT_ZS": "Total Natural Resources Rents (% OF GDP
}

# ============================================
# 1. MISSING VALUES ASSESSMENT
# ============================================
missing_report = []
```

```
for code, label in variable_descriptions.items():
    if code in df.columns:
        missing_count = df[code].isnull().sum()
        missing_pct = (missing_count / len(df)) * 100
        missing_report.append({
            "Variable Code": code,
            "Indicator Description": label,
            "Missing Count": missing_count,
            "Missing (%)": round(missing_pct, 2)
        })

df_missing = pd.DataFrame(missing_report)
print("--- 1. MISSING VALUES ---")
print(df_missing.to_string(index=False))


# ========================================
# 2. INCONSISTENCIES & BOUND CHECKS
# ========================================
print("\n--- 2. INCONSISTENCIES ---")

# Check A: Primary key duplicates (Country + Year)
duplicate_keys = df.duplicated(subset=['country_code', 'year'], k
print(f"Duplicate (Country, Year) records: {duplicate_keys}")

# Check B: Out-of-bounds percentage checks (0% to 100%)
percentage_vars = [
    "SL_TLF_TOTL_FE_ZS",
    "SG_GEN_PARL_ZS",
    "ER_PTD_TOTL_ZS"
]

for code in percentage_vars:
    if code in df.columns:
        label = variable_descriptions[code]
        invalid_rows = df[(df[code] < 0) | (df[code] > 100)]
        print(f"Out of range (0-100%) in '{label}': {len(invalid_

# Check C: Negative values in rent indicators (% of GDP cannot be
rent_vars = ["NY_GDP_MINR_RT_ZS", "NY_GDP_TOTL_RT_ZS"]
for code in rent_vars:
    if code in df.columns:
        label = variable_descriptions[code]
        negative_rows = df[df[code] < 0]
        print(f"Negative values found in '{label}': {len(negative


# ========================================
# 3. OTHER DATA QUALITY ISSUES (OUTLIERS & RANGES)
# ========================================
print("\n--- 3. OUTLIERS AND RANGE SUMMARY ---")

outlier_report = []

for code, label in variable_descriptions.items():
    if code in df.columns:
```

```
        Q1 = df[code].quantile(0.25)
        Q3 = df[code].quantile(0.75)
        IQR = Q3 - Q1

        # Outlier condition using 1.5 * IQR rule
        outliers_count = df[(df[code] < (Q1 - 1.5 * IQR)) | (df[c

        outlier_report.append({
            "Indicator Description": label,
            "Min Value": round(df[code].min(), 2),
            "Max Value": round(df[code].max(), 2),
            "Outliers Count (IQR)": outliers_count
        })

df_outliers = pd.DataFrame(outlier_report)
print(df_outliers.to_string(index=False))
```

```
--- 1. MISSING VALUES ---
    Variable Code                                Indicator Descripti
on  Missing Count  Missing (%)
    GOV_WGI_PV_SC                  Political Stability (SCORE 0-10
0)           4999        47.79
   GOV_WGI_CC_EST       Control of Corruption (governance estimat
e)           5053        48.30
SL_TLF_TOTL_FE_ZS              Female Share of the Labour Force
(%)           3743        35.78
   SG_GEN_PARL_ZS          Women's Share of Parliamentary Seats
(%)           5202        49.73
   ER_PTD_TOTL_ZS Protected Areas (% of total land and marine are
a)           7715        73.75
NY_GDP_MINR_RT_ZS                            Mineral Rents (% of GD
P)           1194        11.41
NY_GDP_TOTL_RT_ZS          Total Natural Resources Rents (% OF GD
P)           1226        11.72

--- 2. INCONSISTENCIES ---
Duplicate (Country, Year) records: 0
Out of range (0–100%) in 'Female Share of the Labour Force (%)':
0 rows
Out of range (0–100%) in 'Women's Share of Parliamentary Seats
(%)': 0 rows
Out of range (0–100%) in 'Protected Areas (% of total land and ma
rine area)': 0 rows
Negative values found in 'Mineral Rents (% of GDP)': 0 rows
Negative values found in 'Total Natural Resources Rents (% OF GD
P)': 0 rows

--- 3. OUTLIERS AND RANGE SUMMARY ---
                           Indicator Description  Min Value  Max
Value  Outliers Count (IQR)
                Political Stability (SCORE 0-100)      12.20
96.49                      4
      Control of Corruption (governance estimate)      -2.10
2.49                       0
            Female Share of the Labour Force (%)       6.78
55.36                    511
         Women's Share of Parliamentary Seats (%)       0.00
63.75                      36
Protected Areas (% of total land and marine area)       0.00
100.00                     71
                         Mineral Rents (% of GDP)       0.00
46.91                    1637
         Total Natural Resources Rents (% OF GDP)       0.00
88.59                    930
```

This table gives us information about our data set, which will help us identify how to narrow down our data for optimal analysis. The current data set has a rather high missing count

for each indicator. This is due to its dimensions being very large covering a wide set of data from decades ago. We therefore considered narrowing down our data to 2013-2023 to enable a more acccurate dataset with less missing values which will be presented in the next section.

Data Filtering

Here we have reduced the temporal scope of our data to between 2013 and 2023 due to the many missing data points from previous years. We therefore have less missing values for each indicator. How we deal with these remaining missing variables will be explained in our exploratory data section.

In [8]:
```
# Filters years from 2013 to 2023 inclusive
df_2013_2023 = df[df['year'].between(2013, 2023)].copy()
```

In [9]:
```
df_2013_2023.head()
```

Out[9]:

	country_code	country	year	ER_PTD_TOTL_ZS	GOV_WGI_CC_EST
27	ABW	Aruba	2013	0.0	1.027719
28	ABW	Aruba	2014	0.0	0.893613
29	ABW	Aruba	2015	0.0	1.159787
30	ABW	Aruba	2016	0.0	1.161127
31	ABW	Aruba	2017	0.1	1.160175

In [10…
```
print(f"\n{countries['country_code'].nunique()} countries/territo
```

```
217 countries/territories when aggregates are removed, 10461 rows
```

Here you can see the head of our final dataset. 217 countries and territories, 7 indicators and limited to 2013-2023.

Data Dictionary

In this section we redefined our variable names into readable outputs.

We then created a final data dictionary in order to understand the demographics of our data better.

Furthermore, we performed an analysis to identify missing values or any outliers in our data. The table shows us a significant decrease in missing values after having restricted the year time frame to 2013-2023. All our indicators have the same number of total data points: 1524.

In [11…

```
variable_descriptions = { #redefining our variable names
    "GOV_WGI_PV_SC": "Political Stability (SCORE 0-100)",
    "GOV_WGI_CC_EST": "Control of Corruption (governance estimate
    "SL_TLF_TOTL_FE_ZS": "Female Share of the Labour Force (%)",
    "SG_GEN_PARL_ZS": "Women's Share of Parliamentary Seats (%)",
    "ER_PTD_TOTL_ZS": "Protected Areas (% of total land and marir
    "NY_GDP_MINR_RT_ZS": "Mineral Rents (% of GDP)",
    "NY_GDP_TOTL_RT_ZS": "Total Natural Resources Rents (% OF GDP
}
```

In [12…

```
from IPython.display import HTML, display
import pandas as pd

# 1. Load dataset and filter panel (2013-2023)
df = pd.read_csv("combined_wdi.csv")
df_panel= df
```

```
# 2. Define model variables
dep_var = "ER_PTD_TOTL_ZS"
indep_vars = [
    "SL_TLF_TOTL_FE_ZS",
    "SG_GEN_PARL_ZS",
    "GOV_WGI_PV_SC",
    "GOV_WGI_CC_EST",
    "NY_GDP_MINR_RT_ZS",
    "NY_GDP_TOTL_RT_ZS",
]

all_vars = ["country_code", "country", "year", dep_var] + indep_v
model_data = df_panel[all_vars].dropna()

# 3. Metadata dictionaries
variable_names = {
    "country_code": "Country Code (ISO-3)",
    "country": "Country Name",
    "year": "Observation Year",
    "ER_PTD_TOTL_ZS": (
        "Terrestrial and Marine Protected Areas (% of total land
    ),
    "SL_TLF_TOTL_FE_ZS": "Female Labor Force Share (% of total la
    "SG_GEN_PARL_ZS": (
        "Women in National Parliaments (% of seats held by women)
    ),
    "GOV_WGI_PV_SC": (
        "Political Stability and Absence of Violence/Terrorism (P
    ),
    "GOV_WGI_CC_EST": "Control of Corruption (WGI Standardized Es
    "NY_GDP_MINR_RT_ZS": "Mineral Rents (% of GDP)",
    "NY_GDP_TOTL_RT_ZS": "Total Natural Resource Rents (% of GDP)
}

variable_roles = {
    "country_code": "Identifier",
    "country": "Identifier",
    "year": "Temporal Index",
    "ER_PTD_TOTL_ZS": "Dependent Variable (DV)",
    "SL_TLF_TOTL_FE_ZS": "Independent Variable (IV)",
    "SG_GEN_PARL_ZS": "Independent Variable (IV)",
    "GOV_WGI_PV_SC": "Independent Variable (IV)",
    "GOV_WGI_CC_EST": "Independent Variable (IV)",
    "NY_GDP_MINR_RT_ZS": "Independent Variable (IV)",
    "NY_GDP_TOTL_RT_ZS": "Independent Variable (IV)",
}

# 4. Construct Data Dictionary rows
data_dict = []
for var in all_vars:
  col_data = model_data[var]
  if pd.api.types.is_numeric_dtype(col_data) and var != "year":
    mean_val = f"{col_data.mean():.2f}"
    std_val = f"{col_data.std():.2f}"
    min_val = f"{col_data.min():.2f}"
    max_val = f"{col_data.max():.2f}"
```

```
  elif var == "year":
    mean_val = "N/A"
    std_val = "N/A"
    min_val = str(int(col_data.min()))
    max_val = str(int(col_data.max()))
  else:
    mean_val = "N/A"
    std_val = "N/A"
    min_val = f"{col_data.nunique()} Unique"
    max_val = "N/A"

  data_dict.append({
      "Code": var,
      "Description": variable_names.get(var, var),
      "Role": variable_roles.get(var, "Control"),
      "Count": f"{len(col_data):,}",
      "Mean": mean_val,
      "Std Dev": std_val,
      "Min": min_val,
      "Max": max_val,
  })

dict_df = pd.DataFrame(data_dict)

# 5. Display as styled HTML Table in Jupyter
html_table = (
    dict_df.style.set_properties(**{"padding": "8px", "text-align
    .set_properties(
        subset=["Code", "Description", "Role"], **{"text-align":
    )
    .set_table_styles([
        {
            "selector": "th",
            "props": [
                ("background-color", "#2B4C7E"),
                ("color", "white"),
                ("font-weight", "bold"),
                ("text-align", "center"),
            ],
        },
        {
            "selector": "tr:nth-of-type(even)",
            "props": [("background-color", "#F4F7FA")],
        },
        {
            "selector": "table",
            "props": [("border", "1px solid #E0E0E0"), ("width",
        },
    ])
    .hide(axis="index")
    .to_html()
)

display(HTML(f"<h3>Master Dataset Data Dictionary</h3>{html_table
```

Master Dataset Data Dictionary

Code	Description	Role	Count	Mean	Std Dev
country_code	Country Code (ISO-3)	Identifier	1,524	N/A	N/A
country	Country Name	Identifier	1,524	N/A	N/A
year	Observation Year	Temporal Index	1,524	N/A	N/A
ER_PTD_TOTL_ZS	Terrestrial and Marine Protected Areas (% of total land area)	Dependent Variable (DV)	1,524	11.95	11.00
SL_TLF_TOTL_FE_ZS	Female Labor Force Share (% of total labor force)	Independent Variable (IV)	1,524	41.41	9.11
SG_GEN_PARL_ZS	Women in National Parliaments (% of seats held by women)	Independent Variable (IV)	1,524	22.20	12.08
GOV_WGI_PV_SC	Political Stability and Absence of Violence/Terrorism (Percentile)	Independent Variable (IV)	1,524	63.53	16.00
GOV_WGI_CC_EST	Control of Corruption (WGI Standardized Estimate)	Independent Variable (IV)	1,524	-0.09	1.00
NY_GDP_MINR_RT_ZS	Mineral Rents (% of GDP)	Independent Variable (IV)	1,524	0.84	2.27
NY_GDP_TOTL_RT_ZS	Total Natural Resource Rents (% of GDP)	Independent Variable (IV)	1,524	6.25	9.11

In [13…

```
# Load dataset and filter for 2013–2023
df = pd.read_csv('combined_wdi_countries_only.csv')
df_2013_2023 = df[df['year'].between(2013, 2023)].copy()

# Define custom variable labels
variable_descriptions = {
    "GOV_WGI_PV_SC": "Political Stability (SCORE 0–100)",
    "GOV_WGI_CC_EST": "Control of Corruption (governance estimate
    "SL_TLF_TOTL_FE_ZS": "Female Share of the Labour Force (%)",
    "SG_GEN_PARL_ZS": "Women's Share of Parliamentary Seats (%)",
    "ER_PTD_TOTL_ZS": "Protected Areas (% of total land and marir
    "NY_GDP_MINR_RT_ZS": "Mineral Rents (% of GDP)",
```

```
    "NY_GDP_TOTL_RT_ZS": "Total Natural Resources Rents (% OF GDP
}

missing_report = []

for code, label in variable_descriptions.items():
    if code in df_2013_2023.columns:
        missing_count = df_2013_2023[code].isnull().sum()
        missing_pct = (missing_count / len(df_2013_2023)) * 100
        missing_report.append({
            "Variable Code": code,
            "Indicator Description": label,
            "Missing Count": missing_count,
            "Missing (%)": round(missing_pct, 2)
        })

df_missing = pd.DataFrame(missing_report)

#For presenting purposes
from IPython.display import display

# Display styled HTML table in Jupyter
display(df_missing)
```

	Variable Code	Indicator Description	Missing Count	Missing (%)
0	GOV_WGI_PV_SC	Political Stability (SCORE 0-100)	121	5.07
1	GOV_WGI_CC_EST	Control of Corruption (governance estimate)	130	5.45
2	SL_TLF_TOTL_FE_ZS	Female Share of the Labour Force (%)	334	13.99
3	SG_GEN_PARL_ZS	Women's Share of Parliamentary Seats (%)	310	12.99
4	ER_PTD_TOTL_ZS	Protected Areas (% of total land and marine area)	67	2.81
5	NY_GDP_MINR_RT_ZS	Mineral Rents (% of GDP)	519	21.74
6	NY_GDP_TOTL_RT_ZS	Total Natural Resources Rents (% OF GDP)	521	21.83

In [14…

```
import pandas as pd
from IPython.display import display

# 1. Define indicators to summarize
indicators = {
    "GOV_WGI_PV_SC":     ("Political stability (0–100)",
    "GOV_WGI_CC_EST":    ("Control of corruption (–2.5 to +2.5)",
    "SL_TLF_TOTL_FE_ZS": ("Women in the workforce (%)",
    "SG_GEN_PARL_ZS":    ("Women in parliament (%)",
    "ER_PTD_TOTL_ZS":    ("Protected nature areas (%)",
```

```
    "NY_GDP_MINR_RT_ZS": ("Income from minerals (% of GDP)",
    "NY_GDP_TOTL_RT_ZS": ("Income from natural resources (% of GD
}

# 2. Build the list of summary dictionary rows
rows = []
for col, (name, meaning) in indicators.items():
    # Take the latest available value for each country
    s = (df.dropna(subset=[col]).sort_values('year')
           .groupby('country').last()[col])
    rows.append({
        'Indicator': name,
        'High number means': meaning,
        'Typical country': round(s.median(), 1),
        'Highest': f"{s.max():.1f} ({s.idxmax()})",
        'Lowest': f"{s.min():.1f} ({s.idxmin()})",
    })

# 3. Create the DataFrame and display it
df_master_audit = pd.DataFrame(rows)
display(df_master_audit)
```

	Indicator	High number means	Typical country	Highest	Lowest
0	Political stability (0-100)	More stable	68.0	93.0 (Andorra)	21.7 (Yemen, Rep.)
1	Control of corruption (-2.5 to +2.5)	Less corruption	-0.2	2.3 (Denmark)	-2.1 (South Sudan)
2	Women in the workforce (%)	More women working	45.6	55.4 (Moldova)	6.8 (Afghanistan)
3	Women in parliament (%)	More women MPs	25.6	63.8 (Rwanda)	0.0 (Oman)
4	Protected nature areas (%)	More land/sea protected	11.6	99.5 (Monaco)	0.0 (Bermuda)
5	Income from minerals (% of GDP)	More reliant on minerals	0.0	28.8 (Congo, Dem. Rep.)	0.0 (Afghanistan)
6	Income from natural resources (% of GDP)	More reliant on resources	1.6	61.0 (Libya)	0.0 (American Samoa)

This table is similar to our data dictionary, however it offers specific country insight and an explanaition of what the values mean.

2.2 Descriptive Statistics

In this section we have described our data by looking at the medians, upper and lower quartiles, and we have found outliers. We have used box plots to visualise the demographics of our dataset.

In [15…

```
#Descriptives Table

# 1. Load dataset and filter for study window (2013–2023)
df = pd.read_csv('combined_wdi_countries_only.csv')
df_2013_2023 = df[df['year'].between(2013, 2023)].copy()

variable_descriptions = {
    "ER_PTD_TOTL_ZS": "Protected Areas (% total land/marine)",
    "GOV_WGI_PV_SC": "Political Stability (Score 0–100)",
    "GOV_WGI_CC_EST": "Control of Corruption (Est.)",
    "SL_TLF_TOTL_FE_ZS": "Female Labor Force Share (%)",
    "SG_GEN_PARL_ZS": "Women in Parliament (%)",
    "NY_GDP_MINR_RT_ZS": "Mineral Rents (% GDP)",
    "NY_GDP_TOTL_RT_ZS": "Natural Resource Rents (% GDP)",
}

cols = list(variable_descriptions.keys())
df_sub = df_2013_2023[cols]

# 2. Compute statistics
stats = pd.DataFrame({
    'Count': df_sub.count(),
    'Mean': df_sub.mean(),
    'Std Dev': df_sub.std(),
    'Min': df_sub.min(),
    'Median': df_sub.median(),
    'Max': df_sub.max(),
    'Skewness': df_sub.skew()
})

# 3. Format index names
stats.index = [variable_descriptions[c] for c in stats.index]
print(stats.round(2))

# 4. Generate visual table graphic
```

```
import statsmodels.api as sm
```

```
                                        Count   Mean  Std Dev    M
in  Median     Max  Skewness
Protected Areas (% total land/marine)   2320  12.70    15.28   0.
00    8.60  100.00      2.76
Political Stability (Score 0-100)       2266  65.37    17.21  15.
90   66.16   96.49     -0.56
Control of Corruption (Est.)            2257  -0.03     1.00  -1.
88   -0.20    2.33      0.44
Female Labor Force Share (%)            2053  41.48     9.23   6.
85   44.96   54.68     -1.63
Women in Parliament (%)                 2077  22.43    12.37   0.
00   20.80   63.75      0.43
Mineral Rents (% GDP)                   1868   0.72     2.13   0.
00    0.00   28.81      6.01
Natural Resource Rents (% GDP)          1866   5.32     8.64   0.
00    1.30   79.43      2.68
```

In [16…

```
import seaborn as sns
from sklearn.preprocessing import StandardScaler

import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load dataset and filter for study period (2013-2023)
df = pd.read_csv('combined_wdi_countries_only.csv')
df_panel = df[df['year'].between(2013, 2023)].copy()

# Set clean aesthetic theme
sns.set_theme(style="whitegrid")

# Define clean indicator labels
labels = {
    "SL_TLF_TOTL_FE_ZS": "Women in Workforce (%)",
    "SG_GEN_PARL_ZS": "Women in Parliament (%)",
    "NY_GDP_MINR_RT_ZS": "Income from Minerals (% GDP)",
    "NY_GDP_TOTL_RT_ZS": "Income from Natural Resources (% GDP)"
}


# ================================================================
# PLOT 1: GENDER EQUALITY (Original Scale: %)
# Indices: Women in Workforce (%) vs Women in Parliament (%)
# ================================================================
fig1, ax1 = plt.subplots(figsize=(8, 4.5), dpi=300)
df_gender = df_panel[['SL_TLF_TOTL_FE_ZS', 'SG_GEN_PARL_ZS']].rer

sns.boxplot(
    data=df_gender,
    orient='h',
    palette=['#8E44AD', '#BB8FCE'],
    ax=ax1,
    fliersize=3
)

ax1.set_title("Gender Equality Indicators Distribution (2013-2023
```

```
ax1.set_xlabel("Percentage (%)", fontsize=10)

plt.tight_layout()
plt.savefig("whisker_gender_equality_final.png", bbox_inches='tig
plt.show()


# ================================================================
# PLOT 2: POLITICAL STABILITY & GOVERNANCE (Standardized Z-Scores
# Indices: Political Stability vs Control of Corruption
# ================================================================
fig2, ax2 = plt.subplots(figsize=(8, 4.5), dpi=300)

# Extract and drop missing values for standardization
df_gov = df_panel[['GOV_WGI_PV_SC', 'GOV_WGI_CC_EST']].dropna().c

# Standardize values to Z-scores (Mean = 0, Std = 1)
scaler = StandardScaler()
scaled_data = scaler.fit_transform(df_gov)

df_gov_scaled = pd.DataFrame(
    scaled_data,
    columns=['Political Stability', 'Control of Corruption']
)

sns.boxplot(
    data=df_gov_scaled,
    orient='h',
    palette=['#2B4C7E', '#41B6C4'],
    ax=ax2,
    fliersize=3
)

ax2.set_title("Political Stability & Corruption Indicators (Stanc
ax2.set_xlabel("Standardized Score (Z-Score)", fontsize=10)

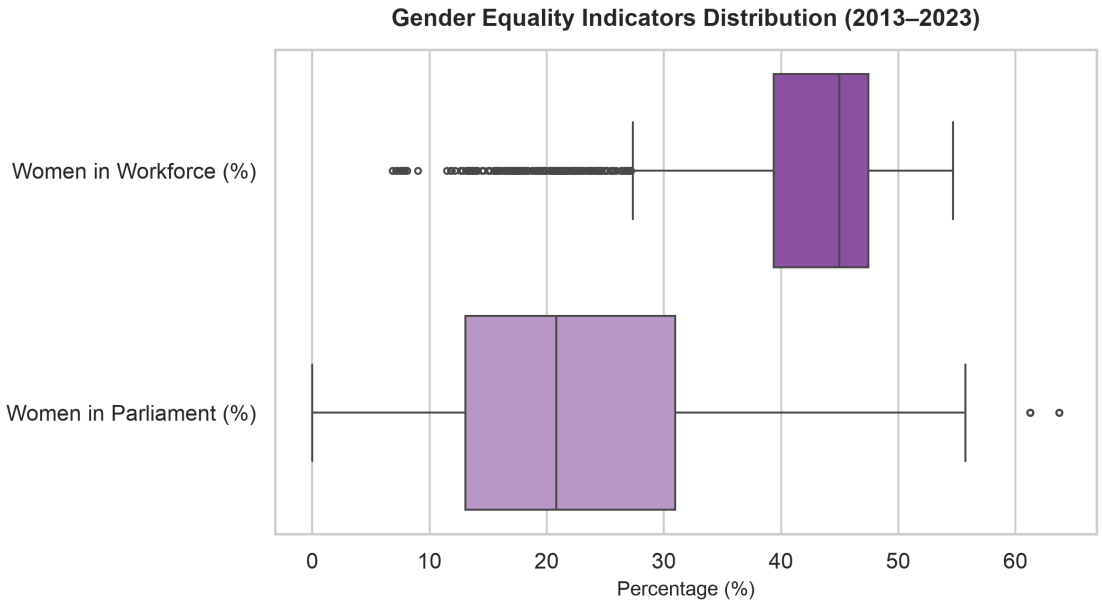
plt.tight_layout()
plt.savefig("whisker_political_stability_zscore_final.png", bbox_
plt.show()

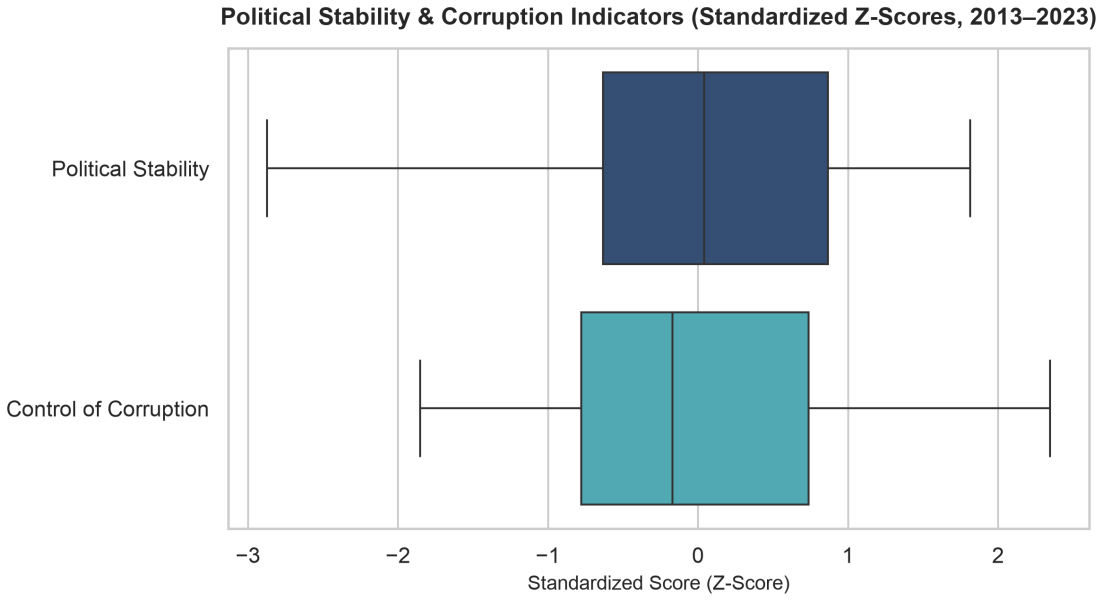

# ================================================================
# PLOT 3: MINERAL & RESOURCE RATES (Original Scale: % of GDP)
# Indices: Income from Minerals (% GDP) vs Income from Natural Re
# ================================================================
fig3, ax3 = plt.subplots(figsize=(8, 4.5), dpi=300)
df_minerals = df_panel[['NY_GDP_MINR_RT_ZS', 'NY_GDP_TOTL_RT_ZS']

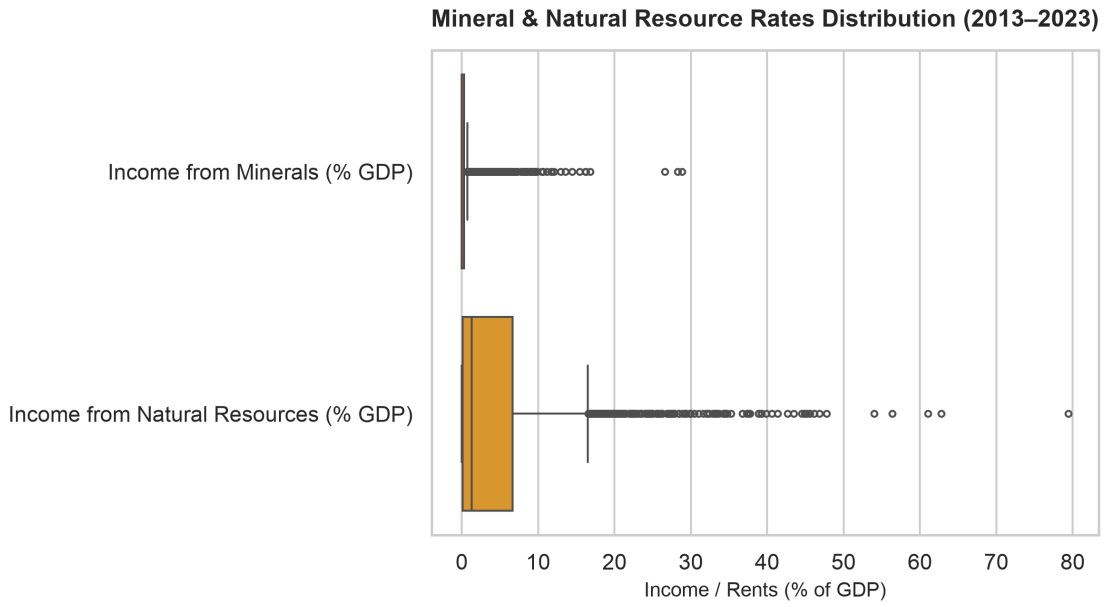
sns.boxplot(
    data=df_minerals,
    orient='h',
    palette=['#E67E22', '#F39C12'],
    ax=ax3,
    fliersize=3
)
```

```
ax3.set_title("Mineral & Natural Resource Rates Distribution (201
ax3.set_xlabel("Income / Rents (% of GDP)", fontsize=10)

plt.tight_layout()
plt.savefig("whisker_mineral_rates_final.png", bbox_inches='tight
plt.show()
```







Interpretation of Whisker Plots:

Female participation in the labour force: This box plot shows a median of 45% of women in the workforce, with a relatively small interquartile range of 8%, which shows fairly limited global variation. Nevertheless, there are many extreme outliers on the lower end.

Women in parliament: There is relatively low representation of women in parliament with a mean of 22.3% and IQR of 18% from 13% to 31%. The country with the highest representation of women in parliament is Rwanda with an average 66.16%.

Political stability: An above average standardized z score median of 0.1. The political stability indicator of the world bank captures perceptions of to what extent it is possible for the government to be destabilised or that authority will be challenged. There is a long negative whisker, which suggests there are values significantly lower than the rest of the data and the median is slightly negatively skewed, suggesting the average is lower than the median.

Corruption: Control of corruption has a lower median than political stability and it is negative, there is also a significant negative skew suggesting that the average is lower than the median. However, the positive whisker is longer, showing that there are more positive outliers than negative.

Mineral rents: The mineral rents box plot has an IQR of 0 and both the upper and lower quartile values are 0, this shows that many of the data points are 0 because the majority of countries do not make money from minerals. Nevertheless, there are outliers and the maximum value is 28.81% of GDP from mineral rents, which is a significant amount of money.

Natural resources rent: Similar to mineral rents the lower quartile value is 0,

> however, the IQR is 6% and there are much higher positive outliers, the maximum value being 79.43% of GDP from natural resources.

2.3 Exploratory Data / Regression

Looking at our empirical setup, we decided that an Ordinary Least Squares (OLS) regression was the most effective choice because it provides clear, directly interpretable coefficients (b). It allows us to see clearly the linear relationship between the dependent variable and its independent variable. We can see if the relationship is positive or negative, and how much each data point affects the prediction of territorial protection. This allows us to quantify the exact relationship between our indicators and protected area coverage.

For missing values, we applied a complete-case analysis. First, we isolated individual countries by removing regional and income aggregates. Next, we restricted our dataset to the 2013–2023 study window where coverage is most complete. Out of 2,387 total country-year observations in the panel, 1,524 complete observations were retained to estimate the final model.

In [17…
```
#Testing running one linear regression
```

In [18…
```
# 1. Load dataset and filter for the 2013-2023 study window
df = pd.read_csv("combined_wdi_countries_only.csv")
df_panel = df[df["year"].between(2013, 2023)].copy()

# 2. Variable mapping dictionary
variable_descriptions = {
    "ER_PTD_TOTL_ZS": "Protected Areas (% total land/marine)",
```

```
    "GOV_WGI_PV_SC": "Political Stability (Score 0-100)",
    "GOV_WGI_CC_EST": "Control of Corruption (Est.)",
    "SL_TLF_TOTL_FE_ZS": "Female Labor Force Share (%)",
    "SG_GEN_PARL_ZS": "Women in Parliament (%)",
    "NY_GDP_MINR_RT_ZS": "Mineral Rents (% GDP)",
    "NY_GDP_TOTL_RT_ZS": "Natural Resource Rents (% GDP)",
}

# 3. Define Dependent Variable (DV) and Independent Variables (I
dep_var = "ER_PTD_TOTL_ZS"
indep_vars = [
    "SL_TLF_TOTL_FE_ZS",  # Female Labor Force Share (%)
    "SG_GEN_PARL_ZS",  # Women in Parliament (%)
    "GOV_WGI_PV_SC",  # Political Stability Score (0-100)
    "GOV_WGI_CC_EST",  # Control of Corruption Estimate (-2.5 to
    "NY_GDP_MINR_RT_ZS",  # Mineral Rents (% GDP)
    "NY_GDP_TOTL_RT_ZS",  # Natural Resource Rents (% GDP)
]

# 4. Drop rows with missing values across all selected model var
model_data = df_panel[[dep_var] + indep_vars].dropna()

# 5. Add constant intercept and fit OLS model with robust HC3 st
X = sm.add_constant(model_data[indep_vars])
y = model_data[dep_var]

model = sm.OLS(y, X).fit(cov_type="HC3")

# ======================================================================
# TABLE 1: MODEL FIT STATISTICS
# ======================================================================
model_fit_table = pd.DataFrame(
    {
        "Metric / Statistic": [
            "Number of Observations (N)",
            "R-squared (R²)",
            "Adjusted R-squared (Adj. R²)",
            "F-statistic",
            "Prob > F (Model p-value)",
            "Log-Likelihood",
            "Akaike Information Criterion (AIC)",
            "Bayesian Information Criterion (BIC)",
        ],
        "Value": [
            f"{int(model.nobs):,}",
            f"{model.rsquared:.4f}",
            f"{model.rsquared_adj:.4f}",
            f"{model.fvalue:.2f}",
            f"{model.f_pvalue:.4e}",
            f"{model.llf:.2f}",
            f"{model.aic:.2f}",
            f"{model.bic:.2f}",
        ],
    }
)
```

```
# ================================================================
# TABLE 2: REGRESSION COEFFICIENTS & HYPOTHESIS TESTING
# ================================================================
coef_labels = ["Constant (Intercept)"] + [
    variable_descriptions[var] for var in indep_vars
]

coefficients_table = pd.DataFrame(
    {
        "Variable Description": coef_labels,
        "Indicator Code": X.columns,
        "Coefficient (β)": model.params.values,
        "Std. Error": model.bse.values,
        "z-statistic": model.tvalues.values,
        "p-value": model.pvalues.values,
        "95% CI Lower": model.conf_int()[0].values,
        "95% CI Upper": model.conf_int()[1].values,
    }
)

# Format numeric outputs for clean print
coefficients_table["Coefficient (β)"] = coefficients_table[
    "Coefficient (β)"
].map("{:.4f}".format)
coefficients_table["Std. Error"] = coefficients_table["Std. Erro
    "{:.4f}".format
)
coefficients_table["z-statistic"] = coefficients_table["z-statis
    "{:.3f}".format
)
coefficients_table["p-value"] = coefficients_table["p-value"].ap
    lambda p: "< 0.001" if p < 0.001 else f"{p:.4f}"
)
coefficients_table["95% CI"] = coefficients_table.apply(
    lambda row: f"[{row['95% CI Lower']:.3f}, {row['95% CI Upper
)

final_coef_table = coefficients_table[
    [
        "Variable Description",
        "Indicator Code",
        "Coefficient (β)",
        "Std. Error",
        "z-statistic",
        "p-value",
        "95% CI",
    ]
]

# Print out results
#For presentation purposes
from IPython.display import display

# Display formatted HTML tables directly in Jupyter
display(model_fit_table)
display(final_coef_table)
```

	Metric / Statistic	Value
0	Number of Observations (N)	1,524
1	R-squared (R^2)	0.1356
2	Adjusted R-squared (Adj. R^2)	0.1321
3	F-statistic	56.13
4	Prob > F (Model p-value)	8.6801e-63
5	Log-Likelihood	-5705.37
6	Akaike Information Criterion (AIC)	11424.73
7	Bayesian Information Criterion (BIC)	11462.03

	Variable Description	Indicator Code	Coefficient (β)	Std. Error	z-statistic	p-value
0	Constant (Intercept)	const	-0.0187	1.4996	-0.012	0.990
1	Female Labor Force Share (%)	SL_TLF_TOTL_FE_ZS	0.2401	0.0258	9.303	< 0.00
2	Women in Parliament (%)	SG_GEN_PARL_ZS	0.1339	0.0214	6.252	< 0.00
3	Political Stability (Score 0-100)	GOV_WGI_PV_SC	-0.0184	0.0219	-0.838	0.402
4	Control of Corruption (Est.)	GOV_WGI_CC_EST	1.9795	0.4313	4.590	< 0.00
5	Mineral Rents (% GDP)	NY_GDP_MINR_RT_ZS	0.5674	0.1341	4.232	< 0.00
6	Natural Resource Rents (% GDP)	NY_GDP_TOTL_RT_ZS	-0.0136	0.0317	-0.428	0.668

In [19…
```
#visualisation
```

In [20…
```
# 1. Load dataset and filter panel (2013–2023)
df = pd.read_csv("combined_wdi_countries_only.csv")
df_panel = df[df["year"].between(2013, 2023)].copy()

# 2. Define dependent and independent variables
dep_var = "ER_PTD_TOTL_ZS"
indep_vars = [
```

```
    "SL_TLF_TOTL_FE_ZS",  # Female Labor Force Share (%)
    "SG_GEN_PARL_ZS",  # Women in Parliament (%)
    "GOV_WGI_PV_SC",  # Political Stability Score
    "GOV_WGI_CC_EST",  # Control of Corruption Estimate
    "NY_GDP_MINR_RT_ZS",  # Mineral Rents (% GDP)
    "NY_GDP_TOTL_RT_ZS",  # Natural Resource Rents (% GDP)
]

# Drop missing values across model variables
model_data = df_panel[[dep_var] + indep_vars].dropna()

# 3. Fit Master OLS Model (HC3 Robust)
X = sm.add_constant(model_data[indep_vars])
y = model_data[dep_var]
model = sm.OLS(y, X).fit(cov_type="HC3")

# 4. Filter for ONLY statistically significant variables (p < 0.
sig_vars = [
    (
        "SL_TLF_TOTL_FE_ZS",
        "Female Labor Force Share (%)",
        "#8E44AD",
    ),  # Purple
    ("SG_GEN_PARL_ZS", "Women in Parliament (%)", "#27AE60"),  #
    ("GOV_WGI_CC_EST", "Control of Corruption Index", "#2B4C7E"
    ("NY_GDP_MINR_RT_ZS", "Mineral Rents (% GDP)", "#E67E22"),
]

# Set visual theme
sns.set_theme(style="whitegrid")

# ==========================================================
# GRAPH 1: 2x2 PARTIAL REGRESSION GRID WITH TRIANGLES
# ==========================================================
fig, axes = plt.subplots(2, 2, figsize=(14, 10), dpi=300)
axes = axes.flatten()

for idx, (var, title, color) in enumerate(sig_vars):
  ax = axes[idx]

  # Plot country-year observations using triangle markers ('^')
  ax.scatter(
      model_data[var],
      model_data[dep_var],
      color=color,
      alpha=0.35,
      marker="^",  # Triangles
      s=30,
      edgecolors="none",
      label="Country-Year Observations",
  )

  # Model partial effect prediction (holding all other predictor
  x_grid = np.linspace(model_data[var].min(), model_data[var].ma
  pred_df = pd.DataFrame({v: [model_data[v].median()] * 100 for
  pred_df[var] = x_grid
```

```
  pred_X = sm.add_constant(pred_df, has_constant="add")[X.columr

  y_pred = model.predict(pred_X)
  ax.plot(
      x_grid,
      y_pred,
      color=color,
      linewidth=2.5,
      label="Master Model Partial Effect",
  )

  ax.set_title(
      title, fontsize=12, fontweight="bold", color="#2C3E50", pa
  )
  ax.set_xlabel(title, fontsize=10)
  ax.set_ylabel("Protected Areas (% total land/marine)", fontsi
  ax.legend(loc="upper left", frameon=True, fontsize=9)

plt.suptitle(
    "Master Regression: Statistically Significant Predictors of
    " Areas",
    fontsize=14,
    fontweight="bold",
    y=0.98,
)
plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.savefig("significant_predictors_grid.png", bbox_inches="tigh
plt.show()

# ============================================================
# GRAPH 2: FOREST PLOT (ONLY SIGNIFICANT VARIABLES, TRIANGLES &
# ============================================================
# Standardize predictors to plot comparative standardized betas
X_std = (
    model_data[indep_vars] - model_data[indep_vars].mean()
) / model_data[indep_vars].std()
X_std = sm.add_constant(X_std)
model_std = sm.OLS(y, X_std).fit(cov_type="HC3")

sig_keys = [v[0] for v in sig_vars]
std_summary = pd.DataFrame({
    "Variable": [v[1] for v in sig_vars],
    "Code": sig_keys,
    "Color": [v[2] for v in sig_vars],
    "Std_Beta": model_std.params[sig_keys].values,
    "CI_Lower": model_std.conf_int().loc[sig_keys, 0].values,
    "CI_Upper": model_std.conf_int().loc[sig_keys, 1].values,
})

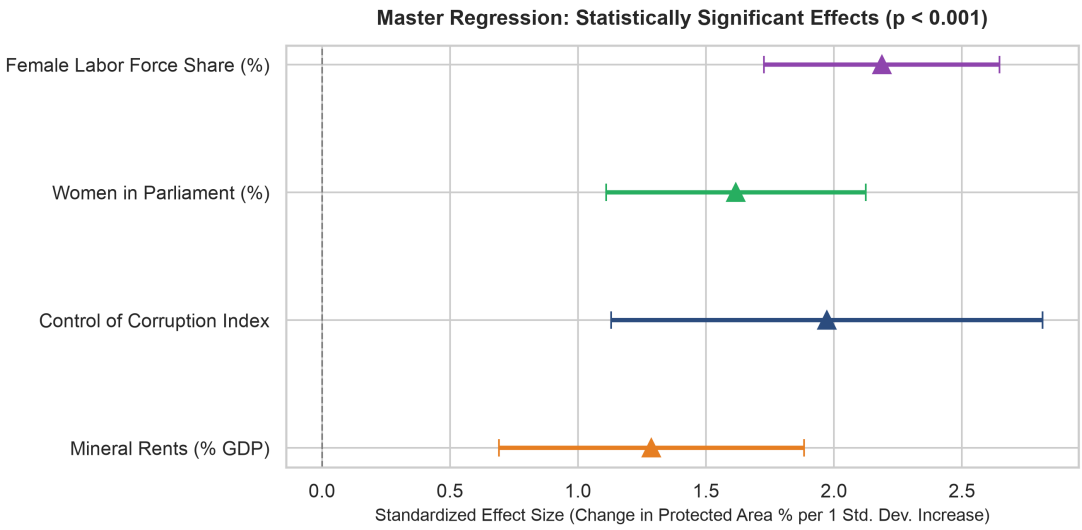
fig_f, ax_f = plt.subplots(figsize=(9, 4.5), dpi=300)

for i, row in std_summary.iterrows():
  ax_f.errorbar(
      x=row["Std_Beta"],
      y=i,
      xerr=[
```

```
            [row["Std_Beta"] - row["CI_Lower"]],
            [row["CI_Upper"] - row["Std_Beta"]],
        ],
        fmt="^",  # Triangle marker
        color=row["Color"],
        ecolor=row["Color"],
        elinewidth=2.5,
        capsize=5,
        markersize=10,
        label=row["Variable"],
    )

ax_f.axvline(x=0, color="gray", linestyle="--", linewidth=1)
ax_f.set_yticks(range(len(std_summary)))
ax_f.set_yticklabels(std_summary["Variable"], fontsize=11)
ax_f.set_xlabel(
    "Standardized Effect Size (Change in Protected Area % per 1
    " Increase)",
    fontsize=10,
)
ax_f.set_title(
    "Master Regression: Statistically Significant Effects (p < 0
    fontsize=12,
    weight="bold",
    pad=12,
)
ax_f.invert_yaxis()
plt.tight_layout()
plt.savefig("significant_forest_plot_triangles.png", bbox_inches
plt.show()
```





Interpretation of Regression Results and Graphs: Our overall model is statistically significant F(6, 1517)=56.13, p<.001 This means that our model witht he specific selected indices could be used as a meaningful framework to measure or predict protected areas and their drivers.
Our linear regression accounts for 13.6% of the variance of national protected areas ($R^2 = 0.136$).

Female Labour Force: Statistically significant (b=0.2401, t=9.30, p<.001). For every 1% increase in female labour force, protected areas increase by 0.24% when all other indices are constant.

Females in Parliament: Statistically significant (b=0.1339, t=6.25, p<.001). For every 1% increase in female parliamentary representation, protected areas increase by 0.13% when all other variables are constant.

Control of Corruption: Statistically significant (b=1.9795, t=4.59, p<.001). For every 1 unit increase in control of corruption, protected areas increase by 1.98% when all other variables are constant.

Political Stability: Not Statistically significant (b=-0.084, t=-0.84, p<.402). There is not statistically meaningful relationship between political stability and protected areas per nation.

Mineral Rents: Statistically significant (b=0.5674, t=4.23, p<.001).

For every 1% increase in mineral rents relative to GDP, protected areas increase by 0.57% when all other variables are constant.

Total Natural Resources: Not Statistically significant (b=-0.0136, t=-0.43, p<.669). There is not statistically meaningful relationship between total natural resources and protected areas per nation.

In [1]:
```
import pandas as pd
import statsmodels.api as sm
import plotly.express as px

# 1. Load Data
df = pd.read_csv("combined_wdi.csv")

AGGREGATES = {
    'AFE','AFW','ARB','CEB','CSS','EAP','EAR','EAS','ECA','ECS
    'FCV','HIC','HPC','IBD','IBT','IDA','IDB','IDX','LAC','LCN
    'LMC','LMY','LTE','MEA','MIC','MNA','NAC','OED','OSS','PRE
    'SAS','SSA','SSF','SST','TEA','TEC','TLA','TMN','TSA','TSS
}

df_c = df[~df['country_code'].isin(AGGREGATES)].copy()

# Fit OLS model on clean panel (2013-2023)
df_panel = df_c[df_c["year"].between(2013, 2023)].copy()
dep_var = "ER_PTD_TOTL_ZS"
indep_vars = [
    "SL_TLF_TOTL_FE_ZS", "SG_GEN_PARL_ZS", "GOV_WGI_PV_SC",
    "GOV_WGI_CC_EST", "NY_GDP_MINR_RT_ZS", "NY_GDP_TOTL_RT_ZS"
]

clean_panel = df_panel[["country_code", "country", "year", dep
X = sm.add_constant(clean_panel[indep_vars])
y = clean_panel[dep_var]
model = sm.OLS(y, X).fit(cov_type="HC3")

# 2. Get LATEST AVAILABLE value up to 2023 for each country (f
df_latest = (
    df_c[df_c["year"] <= 2023]
    .dropna(subset=[dep_var] + indep_vars)
    .sort_values("year")
    .groupby("country_code")
    .last()
    .reset_index()
)

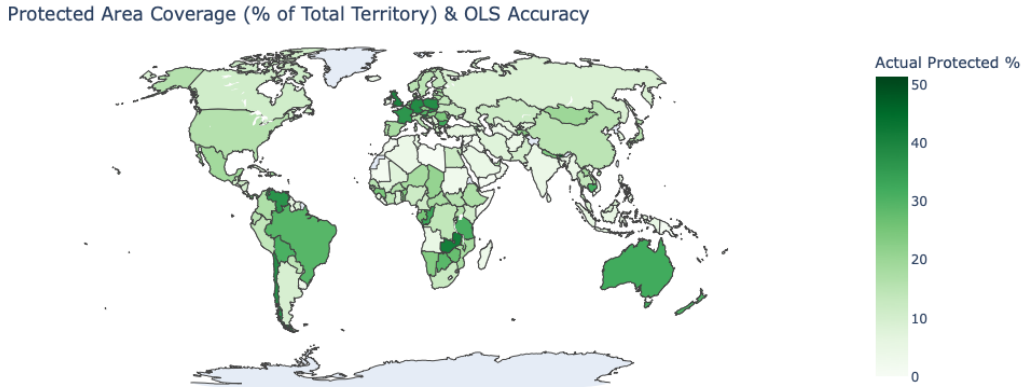
# Predict values & calculate model accuracy
X_latest = sm.add_constant(df_latest[indep_vars])
df_latest["Predicted_Protection"] = model.predict(X_latest).ro
```

```
df_latest["Actual_Protection"] = df_latest[dep_var].round(1)
df_latest["Absolute_Error"] = (df_latest["Actual_Protection"]
df_latest["Model_Accuracy_%"] = (100 - df_latest["Absolute_Err

# 3. Plot Interactive Map with Deep Green Scale
fig = px.choropleth(
    df_latest,
    locations="country_code",
    color="Actual_Protection",
    hover_name="country",
    hover_data={
        "country_code": False,
        "Actual_Protection": ":.1f%",
        "Predicted_Protection": ":.1f%",
        "year": True
    },
    color_continuous_scale="Greens",  # Deeper green for highe
    range_color=[0, df_latest["Actual_Protection"].max()],
    title="Protected Area Coverage (% of Total Territory) & OL
    labels={
        "Actual_Protection": "Actual Protected %",
        "Predicted_Protection": "Model Predicted %",
        "year": "Data Year"
    }
)

fig.update_layout(
    margin={"r": 0, "t": 40, "l": 0, "b": 0},
    geo=dict(showframe=False, showcoastlines=True, projection_
)

fig.show()
```



We were curious to explore the percentage of protected area in the different regions of the world in order to make more sense of our regression results. We therefore made an interactive map that tells us the actual percentage of protected area for the last recorded valid year in each state

and the prediction of our regression model for that same year.

Our model was able to predict some countries better than others, such as the USA with an actual protected area of 15.9% compared to our predicted value of 15.6% in 2021. For other countries such as the United Kingdom our model predicted a percentage of 18% for 2021, however, the real world output was 39.9%.

This discrepancy boils down to our regression only accounting for 13.6% of the protected area results (R^2=0.1356), and only using linear relationships to predict, which is something we will discuss further in our Machine Learning section.

2.4 Zambia, a mineral rents outlier

Given the amount of outliers within the mineral rents indicator, we choose to focus on one of these outliers, Zambia, to illustrate that very few countries have a big part of their GDP coming from mineral rents, while the ones that do can still implement a great level of territorially protected areas.

In [22…

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
```

```
# 1. Load Cleaned Dataset
df = pd.read_csv("combined_wdi_countries_only.csv")

# Filter specifically for 2021 where Zambia's mineral rents h
df_2021 = df[df["year"] == 2021].copy()

# Drop rows missing either variable
df_plot = df_2021.dropna(
    subset=["NY_GDP_MINR_RT_ZS", "ER_PTD_TOTL_ZS"]
).copy()

# Extract Zambia entry
zambia = df_plot[df_plot["country_code"] == "ZMB"].iloc[0]

print("=== ZAMBIA 2021 STATISTICAL HIGHLIGHT ===")
print(f"Mineral Rents (% of GDP): {zambia['NY_GDP_MINR_RT_ZS'
print(f"Protected Area (%):       {zambia['ER_PTD_TOTL_ZS']:.

# 2. Plot Scatter Plot
plt.figure(figsize=(10, 6), dpi=300)
sns.scatterplot(
    data=df_plot,
    x="NY_GDP_MINR_RT_ZS",
    y="ER_PTD_TOTL_ZS",
    color="#94a3b8",
    alpha=0.6,
    s=70,
    edgecolor="none",
)

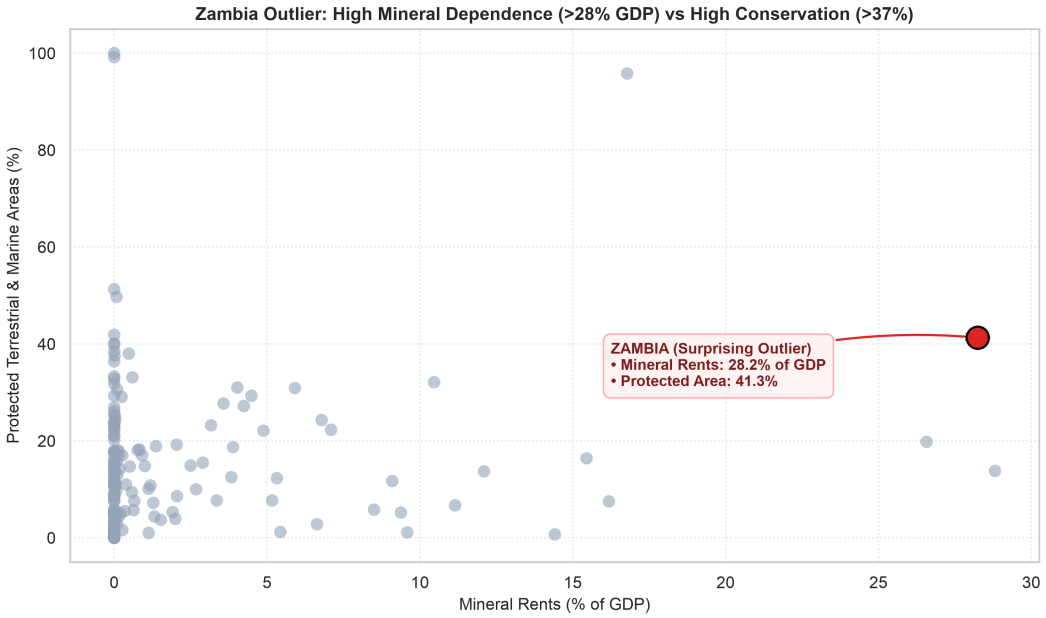
# Highlight Zambia prominently on the far right
plt.scatter(
    zambia["NY_GDP_MINR_RT_ZS"],
    zambia["ER_PTD_TOTL_ZS"],
    color="#dc2626",
    s=220,
    zorder=5,
    edgecolor="black",
    linewidth=1.5,
)

# Callout annotation
plt.annotate(
    f"ZAMBIA (Surprising Outlier)\n• Mineral Rents: {zambia['
    xy=(zambia["NY_GDP_MINR_RT_ZS"], zambia["ER_PTD_TOTL_ZS"]
    xytext=(
        zambia["NY_GDP_MINR_RT_ZS"] - 12,
        zambia["ER_PTD_TOTL_ZS"] - 10,
    ),
    arrowprops=dict(
        arrowstyle="->", color="#dc2626", lw=1.5, connections
    ),
    fontsize=10,
    fontweight="bold",
    color="#7f1d1d",
    bbox=dict(
```

```
        boxstyle="round,pad=0.5",
        facecolor="#fef2f2",
        edgecolor="#fca5a5",
        alpha=0.9,
    ),
)

plt.title(
    "Zambia Outlier: High Mineral Dependence (>28% GDP) vs Hi
    fontsize=12,
    fontweight="bold",
)
plt.xlabel("Mineral Rents (% of GDP)", fontsize=11)
plt.ylabel("Protected Terrestrial & Marine Areas (%)", fontsi
plt.grid(True, linestyle=":", alpha=0.5)
plt.tight_layout()
plt.show()
```

```
=== ZAMBIA 2021 STATISTICAL HIGHLIGHT ===
Mineral Rents (% of GDP): 28.25%
Protected Area (%):       41.30%
```



In [23…

```
# 1. Load data and run master model
df = pd.read_csv("combined_wdi_countries_only.csv")
df_panel = df[df["year"].between(2013, 2023)].copy()

dep_var = "ER_PTD_TOTL_ZS"
indep_vars = [
    "SL_TLF_TOTL_FE_ZS",  # Female Labor Force Share
    "SG_GEN_PARL_ZS",  # Women in Parliament
    "GOV_WGI_PV_SC",  # Political Stability
    "GOV_WGI_CC_EST",  # Control of Corruption
    "NY_GDP_MINR_RT_ZS",  # Mineral Rents
    "NY_GDP_TOTL_RT_ZS",  # Total Resource Rents
]

model_data = df_panel[["country_code", "country", "year", dep

X = sm.add_constant(model_data[indep_vars])
```

```
y = model_data[dep_var]
model = sm.OLS(y, X).fit(cov_type="HC3")


# 2. APA Helper Function for p-values (omits leading zero per
def format_apa_p(p_val):
  if p_val < 0.001:
    return "< .001"
  else:
    return f"= {p_val:.3f}".replace("0.", ".")


# 3. Overall Model Fit Extraction (APA Style)
n_obs = int(model.nobs)
df_model = int(model.df_model)
df_resid = int(model.df_resid)
f_stat = model.fvalue
f_pvalue = model.f_pvalue
r_squared = model.rsquared
adj_r_squared = model.rsquared_adj

# Formatted APA Model Summary Paragraph
apa_model_summary = (
    f"The overall regression model was statistically signific
    f"F({df_model}, {df_resid}) = {f_stat:.2f}, p {format_apa
    f"R² = {r_squared:.3f} (Adjusted R² = {adj_r_squared:.3f}
)

print("=== OVERALL MODEL FIT (APA 7TH) ===")
print(apa_model_summary)
print("\n" + "=" * 50 + "\n")

# 4. Predictor-Level Extraction (APA Style)
apa_rows = []
for var in X.columns:
  b = model.params[var]
  se = model.bse[var]
  t_val = model.tvalues[var]
  p_val = model.pvalues[var]
  ci_lower, ci_upper = model.conf_int().loc[var]

  is_sig = "Significant" if p_val < 0.05 else "Not Significan

  # APA inline reporting string: b = X.XX, SE = X.XX, t(df) =
  apa_string = (
      f"b = {b:.2f}, SE = {se:.2f}, t({df_resid}) = {t_val:.2
      f"p {format_apa_p(p_val)}, 95% CI [{ci_lower:.2f}, {ci_
  )

  apa_rows.append({
      "Variable": var,
      "b": f"{b:.4f}",
      "SE": f"{se:.4f}",
      "t": f"{t_val:.2f}",
      "p": format_apa_p(p_val),
      "Significance (α = .05)": is_sig,
```

```
        "APA Reported String": apa_string,
    })

apa_df = pd.DataFrame(apa_rows)

print("=== INDIVIDUAL PREDICTORS TABLE (APA 7TH) ===")
print(
    apa_df[
        ["Variable", "b", "SE", "t", "p", "Significance (α =
    ].to_string(index=False)
)
```

```
=== OVERALL MODEL FIT (APA 7TH) ===
The overall regression model was statistically significant, F(
6, 1517) = 56.13, p < .001, R² = 0.136 (Adjusted R² = 0.132),
N = 1,524.

==================================================

=== INDIVIDUAL PREDICTORS TABLE (APA 7TH) ===
         Variable       b     SE     t      p Significance (α
= .05)
            const -0.0187 1.4996 -0.01 = .990        Not Signi
ficant
SL_TLF_TOTL_FE_ZS  0.2401 0.0258  9.30 < .001            Signi
ficant
   SG_GEN_PARL_ZS  0.1339 0.0214  6.25 < .001            Signi
ficant
    GOV_WGI_PV_SC -0.0184 0.0219 -0.84 = .402        Not Signi
ficant
   GOV_WGI_CC_EST  1.9795 0.4313  4.59 < .001            Signi
ficant
NY_GDP_MINR_RT_ZS  0.5674 0.1341  4.23 < .001            Signi
ficant
NY_GDP_TOTL_RT_ZS -0.0136 0.0317 -0.43 = .669        Not Signi
ficant
```

In [24…

```
#post - hoc

# 1. Load data and filter study window
df = pd.read_csv("combined_wdi_countries_only.csv")
df_panel = df[df["year"].between(2013, 2023)].copy()

dep_var = "ER_PTD_TOTL_ZS"
indep_vars = [
    "SL_TLF_TOTL_FE_ZS",
    "SG_GEN_PARL_ZS",
    "GOV_WGI_PV_SC",
    "GOV_WGI_CC_EST",
    "NY_GDP_TOTL_RT_ZS",
]

model_data = df_panel[[dep_var] + indep_vars + ["NY_GDP_MINR_

# 2. Apply log-transformation ln(1 + x) to stabilize variance
model_data["log_mineral_rents"] = np.log1p(model_data["NY_GDP
```

```
# 3. Fit OLS with HC3 robust standard errors
X_posthoc = sm.add_constant(
    model_data[indep_vars + ["log_mineral_rents"]]
)
y = model_data[dep_var]

posthoc_model = sm.OLS(y, X_posthoc).fit(cov_type="HC3")

# 4. Display Post-Hoc Results
print("=== POST-HOC LOG-TRANSFORMED REGRESSION ===")
print(posthoc_model.summary())

from IPython.display import HTML, display
```

```
=== POST-HOC LOG-TRANSFORMED REGRESSION ===
                            OLS Regression Results
==============================================================================
Dep. Variable:         ER_PTD_TOTL_ZS   R-squared:                       0.141
Model:                            OLS   Adj. R-squared:                  0.138
Method:                 Least Squares   F-statistic:                     62.69
Date:                Sun, 25 Oct 2026   Prob (F-statistic):           1.25e-69
Time:                        14:12:51   Log-Likelihood:                -5700.3
No. Observations:                1524   AIC:                         1.141e+04
Df Residuals:                    1517   BIC:                         1.145e+04
Df Model:                           6
Covariance Type:                  HC3
==============================================================================
                        coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const                -0.3832      1.488     -0.258      0.797      -3.299       2.533
SL_TLF_TOTL_FE_ZS     0.2359      0.026      9.114      0.000       0.185       0.287
SG_GEN_PARL_ZS        0.1300      0.021      6.099      0.000       0.088       0.172
GOV_WGI_PV_SC        -0.0153      0.022     -0.702      0.483      -0.058       0.027
GOV_WGI_CC_EST        2.0955      0.429      4.880      0.000       1.254       2.937
NY_GDP_TOTL_RT_ZS    -0.0077      0.031     -0.246      0.806      -0.069       0.054
log_mineral_rents     2.6012      0.476      5.470      0.000       1.669       3.533
==============================================================================
Omnibus:                      271.027   Durbin-Watson:                   0.269
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              466.080
Skew:                           1.129   Prob(JB):                     6.19e-102
Kurtosis:                       4.498   Cond. No.                         620.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

3. Discussion and Conclusions of Results

3.1 Discussion

In our project we wanted to investigate the relationship between protected areas and gender equality, political circumstances, rents from natural resources, and minerals. The goal being to determine how adaptations to different areas of policy would affect environmental protection, with the aim to increase the amount of regulated areas, which provide a protective barrier for ecosystems and also an increase to societal well-being.

We began the analysis by creating a condensed data set from the World Bank large dataset as it would be easier to work with. We then limited the range of years to reduce the number of missing values and to make our research more relevant to policy today. We then created a regression to analyse the relationship between 6 indices; women in parliament, female participation in the labour force, mineral rents, natural resources rent, control of corruption and perceptions of political stability and our dependent variable; percentage of protected areas.

3.2 Conclusions and Policy Takeaway

Looking at the results from running our regression, we have come to empirically supported conclusions for each

category of indices:

Gender

According to our regression model, and increased representation of women in parliament and in the labour force positively correlates with increased levels of protected ares. Studies conducted in India and Nepal found that protected forests with committees of higher numbers of women experienced healthier forests, stricter law rule enforcement, and reduced illegal logging (Agarwal, 2009). This can be compared with our results, like the women in the committies, are women in parliament more strict on laws for protected areas? Kahsay et al. (2021) found that female-led projects on forest management in Ethipoia directly linked to improved forest canopy compared to groups without female leadership. This raises questions of women possibly offering alternative perspectives on forest conservation and protection to what has been proposed by men for centuries?

Politics

Our analysis shows that there is a statistically significant connection between control of corruption and amount of protected areas, in fact it is the strongest correlation. This we expected. According to the book, Prisoners of Geography, in Latin America, corruption is one of the biggest enabling factors in environmental destruction. For example, in illegal mining of minerals or deforestation and logging. Further research by Rosana and Magaly shows that at a national level, legislation can be

influenced by bribes or legislators individual intrests. For example, in Peru there is significant risk of this in the drafting of mining laws. An unexpected result was the lack of connection between political stablity and environmental protection. We anticipated that the higher the political stability, the easier it would be to establish protected areas but our data does not support this.

Natural resources

Looking at the natural resource indicators we see some interesting results. First and foremost, we can draw the conclusion that natural resources rents as a percentage of total gdp is not a good predictor for how much of a territory's total area is protected, when controlled for all other indicators previously mentioned. This was an unpredicted outcome, but we can draw the conclusion that factors like political stability and lack of corruption effectively moderate the assumption that an economic dependency on natural resources would lead to a decrease in protected areas. Another factor for this lack of correlation can also be that some of the world's largest natural resource producers, like the United States, Canada, China and Brazil have highly diversified economies. For example, the United States is the largest producer in the world of natural gas, but natural gas rents as a share of GDP was only 0.36 (World Bank, 2026). Ergo, some major producers of natural resources are less reliant on the industries as a whole, and this may cause a moderation effect on the correlation.
For mineral rents we see a positive correlation between mineral rents as a percentage of GDP and

total protected areas, but as previously mentioned mineral rents is a skewed statistic, where very few countries have it as a big part of their GDP. Following this, some of the most mineral heavy countries have a large rate of protected areas, which is a surprising find. Countries such as Zambia, Chile and Australia all have more than 10% of GDP come from mineral rents, but all countries have reached the 30x30 goal of protected areas. Zambia is the most extreme of these countries, where mineral rents were 28,25% of GDP in 2021 , and the total protected area was 41,3% of total territorial area. Finally, mineral extraction is somewhat “efficient” when it comes to the area it uses. A mine often goes down into the ground, and uses less area than industries such as agriculture. Therefore, a large area can be protected at the same time as a high degree of mining may be present.

Given the findings, we recommend that policy makers and NGO´s take the a wider scope when looking at how to achieve the 30x30 goal. Focusing solely on large natural resource rents is not an efficient way to recognise territories with potential for territorial protection. Taking a more holistic approach would be more efficient. Furthermore, research in the future should focus on even more economic factors, such as agriculture and marine industries.

Answering our Questions?:

R1: Is there a relationship between the quantity of protected areas and equal gender employment? YES

R2: Is there a relationship between women's rights performance and the quantity of protected areas? YES

R3: Do increased levels of corruption limit environmental protection? YES

R4: Does political stability encourage more environmental protection? NO%

R5: Does a natural resource based economy lead to less prevalence of protected areas? NO

In [25…

```
import statsmodels.api as sm

# 1. Load dataset and filter for panel study window (2013-2
df = pd.read_csv("combined_wdi_countries_only.csv")
df_panel = df[df["year"].between(2013, 2023)].copy()

# 2. Define model variables
dep_var = "ER_PTD_TOTL_ZS"
indep_vars = [
    "SL_TLF_TOTL_FE_ZS",  # Female Labor Force Share (%)
    "SG_GEN_PARL_ZS",  # Women in Parliament (%)
    "GOV_WGI_PV_SC",  # Political Stability Score
    "GOV_WGI_CC_EST",  # Control of Corruption Index
    "NY_GDP_MINR_RT_ZS",  # Mineral Rents (% GDP)
    "NY_GDP_TOTL_RT_ZS",  # Natural Resource Rents (% GDP)
]

# 3. Clean complete cases for model estimation
model_data = df_panel[
    ["country_code", "country", "year", dep_var] + indep_va
].dropna()

# 4. Fit the Master OLS Model (HC3 Robust)
X = sm.add_constant(model_data[indep_vars])
y = model_data[dep_var]
model = sm.OLS(y, X).fit(cov_type="HC3")

# 5. Calculate model predictions and the Protection Gap (Re
model_data["Predicted_Protection"] = model.predict(X)
model_data["Protection_Gap"] = (
    model_data[dep_var] - model_data["Predicted_Protection"
)

# 6. Aggregate across the panel window by country
country_summary = (
    model_data.groupby(["country_code", "country"])
    .agg(
        Actual_Protection=(dep_var, "mean"),
        Predicted_Protection=("Predicted_Protection", "mean
```

```
        Protection_Gap=("Protection_Gap", "mean"),
        Control_of_Corruption=("GOV_WGI_CC_EST", "mean"),
        Women_in_Parliament=("SG_GEN_PARL_ZS", "mean"),
        Female_Labor_Force=("SL_TLF_TOTL_FE_ZS", "mean"),
    )
    .reset_index()
)

# 7. Extract top 3 countries with the largest negative Prot
top3_countries = country_summary.sort_values("Protection_Ga


from IPython.display import display

# Format the extracted top 3 candidates into a styled Datal
top3_formatted = pd.DataFrame({
    "Country": top3_countries["country"],
    "Actual Protection (%)": top3_countries["Actual_Protec
        "{:.2f}%".format
    ),
    "Model Predicted Protection (%)": top3_countries[
        "Predicted_Protection"
    ].map("{:.2f}%".format),
    "Protection Gap (Residual)": top3_countries["Protectio
        "{:.2f}%".format
    ),
    "Control of Corruption Score": top3_countries["Control_
        "{:.2f}".format
    ),
    "Women in Parliament (%)": top3_countries["Women_in_Pa
        "{:.1f}%".format
    ),
})

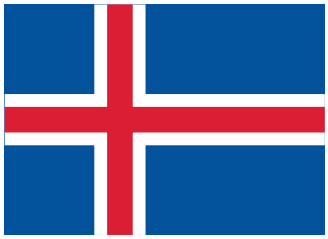
# Display as an interactive Jupyter Notebook table
display(top3_formatted)
```

	Country	Actual Protection (%)	Model Predicted Protection (%)	Protection Gap (Residual)	Control of Corruption Score	Wome Parlian
75	Iceland	2.56%	18.76%	-16.20%	1.88	4(
22	Barbados	0.00%	15.15%	-15.15%	1.17	1
37	Cabo Verde	0.00%	13.74%	-13.74%	0.84	2

Delving Deeper:

Iceland:

Looking at Iceland's output variables (based on our dataset) it



currently has 2.56% of protected marine and terrestrial area, which is amongst the lowest in our data set. Seeing that Iceland has a high control of corruption and large female representatives in parliament, both of which are key drivers of increased protective areas, they have the capacity to prioritise and formalise protected areas in accordance with SDG goals.

Barbados:



Barbados goes against our predicted model. We would assume that a good chunk of their territory would be protected, but this is not the case. Looking at Barbados we see that their economy is driven by cruises and tourists, and the island is experiencing a high level of population density, driven further by cruise guests on day-trips to the island. The lack of available space on the island and the prevalence of marine economies has hampered the development of protected areas. Add to that the fact that Barbados´ exclusive economic zone (ocean) is 400 times larger than the terrestrial area of the island, and it becomes difficult for the state to divert resources and time to effectively protect and police a busy shipping lane outside the island. Our policy suggestion aligns with an already ongoing development in Barbados, the Barbados Marine Spatial Plan, where economic incentives, local business people and a goal of reaching the 30x30 plan is undergoing, and projected to be ready by 2027. Furthermore, we would recommend that international organizations such as the U.N. could aid in targeted funding of marine protection to small island nations in order to lessen the load on resources needed by the states.

Cabo Verde:



In our dataset from the World Bank, Cabo Verde is recorded as having a nationally protected

area of 0.00%, however
according to protectedplanet, the actual current percentage of protected areas is 17.62%. This is because the World Bank dataset has a strict indicator definition, which is that only areas formally reported to the World Datebase on Protected Areas are included. However, what is intresting is that our regression model predicts that Cabo Verde should have a level of around 13.74%, which is fairly similar to the actual number. This suggests that governments with more female representation in parliament and less corruption will create more protected areas, as it suggests there is significance between the variables we have chosen and increase in protected area. Cabo Verde has 0.00% mineral rents, so it isn't a useful study in making conclusions about mineral rents policy and increasing the amount of protected areas. However, it also points to a limitation in our dataset, as it suggests that there is a difference between the data we have used to create our regression and the reality, so our results may have been changed if we used a different dataset.

4. Limitations to our Study

An important point to note about the data is that the statistic does not include the unpaid labour of women raising kids, cleaning the house and familial administration, hence there are women working who are not included.

While our study offers key empirical insights into the drivers of territorial

protection, several limitations must be acknowledged.

First, our model relies on a concise set of seven World Bank indicators, not a lot of indicators in the grand scheme of things. Furthermore, our dataset may fail to capture informal, indigenous, or community-managed conservation areas due to strict reporting protocols. This creates notable data gaps for nations like Cabo Verde, where active coastal protection is not fully reflected in official percentage metrics.

Second, our natural resource metrics focus on gross rents as a percentage of GDP, which omits economic factors such as sector-specific employment, value added to the domestic economy, and competing industries like agricultural or marine.

Third, percentage-based targets like the 30x30 goal treat all landmasses as equal, masking vast physical differences where protecting 30% of a land-abundant nation

like the United States represents millions of square kilometers, but 100% of a microstate like Luxembourg yields much less.

Finally, this project was made within a short period of time, resulting in less research than other similar projects, a major reliance on AI for statistical analysis and coding, and a less polished editing process than usual.

Reflection Question for Class:

Given that protected areas are difficult to remove, and are stuck at a relatively static percentage, does this impact our findings in any way?

5. Machine Learning Attempt

Explaining our Model

We were curious to explore our data further as we were surprised by some of the results from our linear regression model. We were surprised that the Naural Resource Rents alongside Political Stability were not stitistically signifacnt indicators, so we decided to explore the relationship between variables further.

> Unlike linear regression which only measures the relationship between two variables, our random forest regressor enables us to measure the non-linear relationships and interactions between our variables.
>
> Our model builds a forest of 100 decision trees.
>
> It is then able to look at random branches of our country-year cases, splitting it into different paths.
>
> So **in order to predict a country's protected area percentage, our forest regressor passes the country's indicators through all the different decision trees and averages final predictions.**

In [26…

```
#Machine learning
```

In [27…

```
import numpy as np
import pandas as pd
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_scor
from sklearn.model_selection import train_test_split

# 1. Load data and filter panel (2013-2023)
df = pd.read_csv("combined_wdi_countries_only.csv")
df_panel = df[df["year"].between(2013, 2023)].copy()

dep_var = "ER_PTD_TOTL_ZS"  # Target variable
indep_vars = [
    "SL_TLF_TOTL_FE_ZS",  # Female Labor Force Share
    "SG_GEN_PARL_ZS",  # Women in Parliament
    "GOV_WGI_PV_SC",  # Political Stability
    "GOV_WGI_CC_EST",  # Control of Corruption
    "NY_GDP_MINR_RT_ZS",  # Mineral Rents
    "NY_GDP_TOTL_RT_ZS",  # Total Resource Rents
]

# 2. Clean missing values and isolate X and y
model_data = df_panel[[dep_var] + indep_vars].dropna()
X = model_data[indep_vars]
y = model_data[dep_var]

# 3. Train-Test Split (80% Train, 20% Test)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

# 4. Initialize and fit the Random Forest Model
model = RandomForestRegressor(n_estimators=100, random_
model.fit(X_train, y_train)
```

```
# 5. Print evaluation metrics
y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
print(f"Training set size: {len(X_train)} | Test set si
print(f"Test R² Accuracy:  {r2:.2%}")
print(f"Test RMSE Error:   {rmse:.2f}%")
```

```
Training set size: 1219 | Test set size: 305
Test R² Accuracy:  72.76%
Test RMSE Error:   5.57%
```

Understanding the Results - Performance and Power

Firstly comparing the predictive power of each analysis/model:

Regression Analysis:

R^2 = 13.6%, capturing the baseline linear trends across variables

Random Forest Regressor:

R^2 = 72.76%, accurately predicting protected coverage on unseen data with average error of ±5.57% (RMSE=5.57%).

In [28…

```
from IPython.display import HTML, display
import pandas as pd

# 1. Extract feature importances from the trained model
importances = model.feature_importances_

# 2. Build feature importance DataFrame
feature_importance_df = pd.DataFrame(
    {"Indicator": X.columns, "Importance_Score": import
).sort_values(by="Importance_Score", ascending=False)

# 3. Create clean display labels
label_mapping = {
    "SL_TLF_TOTL_FE_ZS": "Female Labor Force Share",
    "SG_GEN_PARL_ZS": "Women in Parliament",
    "NY_GDP_TOTL_RT_ZS": "Total Natural Resource Rents'
    "GOV_WGI_PV_SC": "Political Stability",
    "GOV_WGI_CC_EST": "Control of Corruption",
    "NY_GDP_MINR_RT_ZS": "Mineral Rents",
}

# 4. Format DataFrame for display
```

```
feature_importance_df["Indicator Name"] = feature_impor
    "Indicator"
].map(label_mapping)
feature_importance_df["Importance Weight"] = feature_im
    "Importance_Score"
].map("{:.4f}".format)
feature_importance_df["Importance (%)"] = (
    feature_importance_df["Importance_Score"] * 100
).map("{:.2f}%".format)

table_display = feature_importance_df[
    ["Indicator Name", "Importance Weight", "Importance
]

# 5. Apply clean navy and grey styling
html_table = (
    table_display.style.set_properties(
        **{"padding": "8px", "text-align": "right"}
    )
    .set_properties(subset=["Indicator Name"], **{"text
    .set_table_styles([
        {
            "selector": "th",
            "props": [
                ("background-color", "#2B4C7E"),
                ("color", "white"),
                ("font-weight", "bold"),
                ("text-align", "center"),
            ],
        },
        {
            "selector": "tr:nth-of-type(even)",
            "props": [("background-color", "#F4F7FA")],
        },
        {
            "selector": "table",
            "props": [("border", "1px solid #E0E0E0"),
        },
    ])
    .hide(axis="index")
    .to_html()
)

display(
    HTML(f"<h3>Random Forest Feature Importance Table</
)
```

Random Forest Feature Importance Table

Indicator Name	Importance Weight	Importance (%)
Female Labor Force Share	0.2513	25.13%
Women in Parliament	0.1861	18.61%
Total Natural Resource Rents	0.1715	17.15%
Political Stability	0.1472	14.72%
Control of Corruption	0.1327	13.27%
Mineral Rents	0.1112	11.12%

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```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

# 1. Extract feature importances from the trained model
importances = model.feature_importances_

# 2. Build feature importance DataFrame
feature_importance_df = pd.DataFrame(
    {"Indicator": X.columns, "Importance_Score": import
).sort_values(by="Importance_Score", ascending=False)

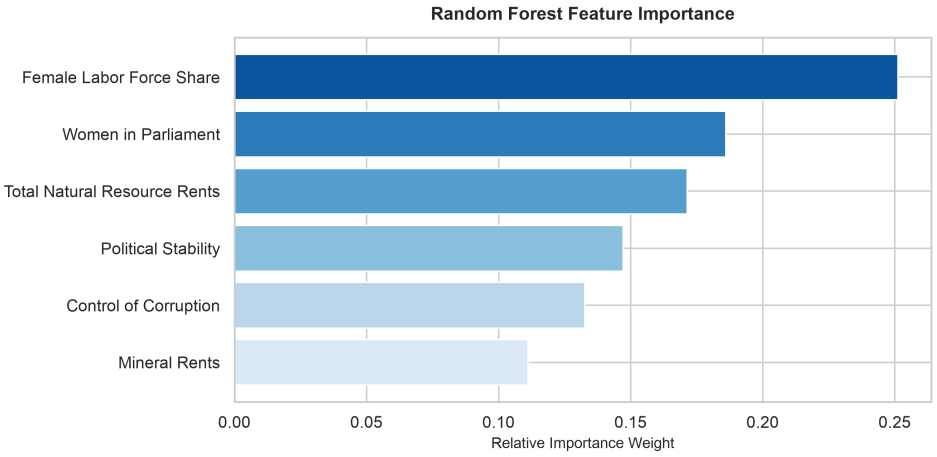
# 3. Create clean display labels
label_mapping = {
    "SL_TLF_TOTL_FE_ZS": "Female Labor Force Share",
    "SG_GEN_PARL_ZS": "Women in Parliament",
    "NY_GDP_TOTL_RT_ZS": "Total Natural Resource Rents'
    "GOV_WGI_PV_SC": "Political Stability",
    "GOV_WGI_CC_EST": "Control of Corruption",
    "NY_GDP_MINR_RT_ZS": "Mineral Rents",
}

feature_importance_df["Display_Name"] = feature_importa
    label_mapping
)

# 4. Plot Horizontal Bar Chart using Matplotlib (warnir
plt.figure(figsize=(9, 4.5), dpi=300)
colors = sns.color_palette("Blues_r", len(feature_impor

plt.barh(
    y=feature_importance_df["Display_Name"],
    width=feature_importance_df["Importance_Score"],
    color=colors,
)
plt.gca().invert_yaxis()  # Highest importance at top
```

```
plt.title(
    "Random Forest Feature Importance", fontsize=12, fo
)
plt.xlabel("Relative Importance Weight", fontsize=10)
plt.tight_layout()
plt.show()
```



Understanding the Results - Each Indicator

According to our machine model, which is able to identify patterns and interactions between variables, **Female Labour Force Share** seems to have the highest relative importance of **25.06%** and is set to be the strongest overall predictor of protected area coverage. This indicates that women's presence in labour force affects total percentage of protected areas both in a linear (as seen in our regression model) and non-linear, indirect manner.

Second comes **Women in Parliament** at **18.67%**, suggesting that female political voices stand a critical role in shaping conservation policy. As seen in India, Nepal and Ethopia in earlier examples, and in our regression, and machine learning model, women seem to have a significant impact on the conservation of the environment.

While **Total Natural Resource Rents** was statistically insignificant in our linear regression, it contributed to **17.17%** prediction accuracy in our machine model. This suggests that resource dependence affects protected area coverage throguh a more complex, **NON-LINEAR** threshold dynamic. Its interaction with our other

variables indirectly influence protection areas rather than through a direct relationship.

The same principle applies to our **Political Stability** indicator.

Control of Corruption was at 13.34% and **Mineral Rents** was at 11.07% which is relatively low compared to their linear regression outcome. This might suggest that they are stronger as independent indices than indirect drivers.

Conclusion

This project investigated how gender equality, governance, and natural resource dependence relate to protected-area coverage across countries between 2013 and 2023. We found several important relationships in the dataset such as: women's labour force participation, women's parliamentary representation, and control of corruption were positively associated with protected-area coverage, which gives quantitative evidence to former qualitative studies. Political stability was not statistically significant, while mineral rents showed a positive association and total natural resource rents were not significant in the linear regression.

Comparing models provided further insight. The linear regression explained approximately 13.6% of the variation, whereas the random forest achieved an (R^2) of 73.1% and an RMSE of 5.53 percentage points. This suggests that nonlinear relationships may be important. Both models highlighted women's participation as an important factor, although importance does not establish causation, and other factors may be even more important. This is something researchers in the future can look deeper into.

The country comparisons also identified potential gaps between observed and predicted protection, including Iceland, Barbados, and Cabo Verde. However, discrepancies

between World Bank and Protected Planet data, particularly for Cabo Verde, demonstrate the importance of reliable and consistent reporting, as well as holistic approaches when designing policy aimed at the 30x30 goal.

Overall, the project contributes evidence that protected-area coverage is associated with interconnected social, political, and economic factors. It suggests that conservation policies, including efforts towards the 30x30 target, should consider gender equality and governance alongside economic conditions. Future research could incorporate even more environmental and socioeconomic indicators, examine changes over time, and investigate regional differences. Although the analysis cannot establish causation, it provides a useful foundation for identifying relationships, challenging assumptions, and guiding further research into effective conservation policy.

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